

You have been provided with data from a company which has many departments. The data shows employee related information and their experience details. You are required to use this data for analysis based on the following guidelines:

## 1. Data Cleaning:

❖ **Check and remove blank values from the dataset.**

The screenshot shows the Microsoft Excel interface. The ribbon is set to the 'Table Design' tab, which includes options for 'Conditional Formatting', 'Format as Table', and 'Cell Styles'. The 'Home' tab is also visible, showing options for 'Clipboard', 'Font', 'Alignment', 'Number', and 'Styles'. The data table is displayed below the ribbon, with columns labeled A through L. The first row of the table is highlighted in orange and contains the following headers: 'EMP ID', 'Age', 'DailyRate (\$)', 'Department', 'Education, Gender', 'HourlyR', 'JobInvo', 'JobLevel', 'Prefix Job R', 'JobSatisfaction', 'MaritalStatu', and 'NumCompani'. The subsequent rows (1453 to 1469) contain data, with the 'Prefix Job R' column consistently showing '#N/A'.

|      | A      | B   | C              | D          | E                 | F       | G       | H        | I            | J               | K            | L          |
|------|--------|-----|----------------|------------|-------------------|---------|---------|----------|--------------|-----------------|--------------|------------|
|      | EMP ID | Age | DailyRate (\$) | Department | Education, Gender | HourlyR | JobInvo | JobLevel | Prefix Job R | JobSatisfaction | MaritalStatu | NumCompani |
| 1453 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1454 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1455 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1456 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1457 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1458 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1459 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1460 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1461 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1462 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1463 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1464 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1465 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1466 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1467 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1468 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |
| 1469 |        |     |                |            |                   |         |         |          | #N/A         |                 |              |            |

❖ 1483 Rows Before Removing the Blank Rows.

**After removing:**

AutoSave Employee Data

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design

Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins

E27  Marketing,Female

|      | EMP ID  | Age | DailyRate (\$) | Department             | Education, Gender       | HourlyRate | JobInvol | JobLevel | Prefix, Job Role | JobSatisfaction | MaritalStatus | NumCompanies |
|------|---------|-----|----------------|------------------------|-------------------------|------------|----------|----------|------------------|-----------------|---------------|--------------|
| 1448 | EM00982 | 51  | 1405           | Research & Development | Technical Degree,Female | 82         | 2        | 4        | MDManufacturi    | 2               | Single        |              |
| 1449 | EM00983 | 18  | 1124           | Research & Development | Life Sciences,Female    | 97         | 3        | 1        | LTLaboratory Te  | 4               | Single        |              |
| 1450 | EM00984 | 35  | 817            | Research & Development | Medical,Female          | 60         | 2        | 2        | LTLaboratory Te  | 4               | Married       |              |
| 1451 | EM00985 | 27  | 793            | Sales                  | Life Sciences,Male      | 43         | 1        | 2        | SESales Executi  | 4               | Single        |              |
| 1452 | EM00986 | 55  | 267            | Sales                  | Marketing,Male          | 85         | 4        | 4        | SESales Executi  | 3               | Single        |              |
| 1453 | EM00987 | 56  | 1369           | Research & Development | Life Sciences,Male      | 68         | 3        | 4        | MDManufacturi    | 2               | Married       |              |
| 1454 | EM00988 | 34  | 999            | Research & Development | Technical Degree,Female | 92         | 2        | 1        | RSResearch Sci   | 3               | Divorced      |              |
| 1455 | EM00989 | 40  | 1202           | Research & Development | Medical,Female          | 89         | 4        | 2        | HCHealthcare F   | 3               | Divorced      |              |
| 1456 | EM00990 | 30  | 288            | Research & Development | Life Sciences,Male      | 99         | 2        | 2        | HCHealthcare F   | 4               | Married       |              |
| 1457 | EM00990 | 34  | 285            | Research & Development | Medical,Male            | 86         | 3        | 2        | LTLaboratory Te  | 3               | Married       |              |
| 1458 | EM00991 | 31  | 703            | Sales                  | Life Sciences,Female    | 90         | 2        | 1        | SRSales Repres   | 4               | Single        |              |
| 1459 | EM00992 | 35  | 662            | Sales                  | Marketing,Female        | 67         | 3        | 2        | SESales Executi  | 3               | Married       |              |
| 1460 | EM00993 | 38  | 693            | Research & Development | Life Sciences,Male      | 57         | 4        | 1        | RSResearch Sci   | 3               | Divorced      |              |
| 1461 | EM00994 | 34  | 404            | Research & Development | Technical Degree,Female | 98         | 3        | 2        | HCHealthcare F   | 4               | Single        |              |
| 1462 | EM00995 | 28  | 736            | Sales                  | Life Sciences,Male      | 48         | 2        | 2        | SESales Executi  | 1               | Married       |              |
| 1463 | EM00996 | 31  | 330            | Research & Development | Medical,Male            | 98         | 3        | 2        | MDManufacturi    | 3               | Married       |              |
| 1464 | EM00997 | 39  | 1498           | Sales                  | Life Sciences,Male      | 44         | 2        | 2        | SESales Executi  | 4               | Married       |              |
| 1465 | EM00998 | 51  | 541            | Sales                  | Marketing,Male          | 52         | 3        | 3        | SESales Executi  | 2               | Married       |              |
| 1466 | EM00999 | 41  | 1200           | Research & Development | Life Sciences,Female    | 75         | 3        | 2        | RSResearch Sci   | 4               | Divorced      |              |

❖ 1466 Rows After removing the Blank Rows.

## ❖ Check if any duplicate rows are present and delete them.

AutoSave Off Employee Data

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design

Clipboard Font Alignment Number Styles Cells Editing Sensitivity

A2 EM00131

| EMPID   | Age | DailyRate (\$) | Department             | Education, Gender     | HourlyR | JobInvc | JobLevel | Prefix Job N    | JobSatisfactio | MaritalStat | NumCompaniesW | OverTime |
|---------|-----|----------------|------------------------|-----------------------|---------|---------|----------|-----------------|----------------|-------------|---------------|----------|
| EM00131 | 30  | 1312           | Research & Development | Life Sciences,Male    | 96      | 1       | 1        | RSResearch Sc   | 3              | Divorced    | 1             | No       |
| EM00131 | 30  | 1312           | Research & Development | Life Sciences,Male    | 96      | 1       | 1        | RSResearch Sc   | 3              | Divorced    | 1             | No       |
| EM00132 | 33  | 750            | Sales                  | Marketing,Male        | 95      | 3       | 2        | SESales Execut  | 2              | Married     | 0             | No       |
| EM00132 | 33  | 750            | Sales                  | Marketing,Male        | 95      | 3       | 2        | SESales Execut  | 2              | Married     | 0             | No       |
| EM00133 | 56  | 441            | Research & Development | Life Sciences,Female  | 72      | 3       | 1        | RSResearch Sc   | 2              | Married     | 9             | Yes      |
| EM00133 | 56  | 441            | Research & Development | Life Sciences,Female  | 72      | 3       | 1        | RSResearch Sc   | 2              | Married     | 9             | Yes      |
| EM00134 | 51  | 684            | Research & Development | Life Sciences,Male    | 51      | 3       | 5        | RDRResearch Di  | 3              | Single      | 7             | No       |
| EM00134 | 51  | 684            | Research & Development | Life Sciences,Male    | 51      | 3       | 5        | RDRResearch Di  | 3              | Single      | 7             | No       |
| EM00135 | 31  | 249            | Sales                  | Life Sciences,Male    | 76      | 1       | 2        | SESales Execut  | 3              | Married     | 4             | Yes      |
| EM00135 | 31  | 249            | Sales                  | Life Sciences,Male    | 76      | 1       | 2        | SESales Execut  | 3              | Married     | 4             | Yes      |
| EM00136 | 26  | 841            | Research & Development | Other,Female          | 46      | 2       | 1        | RSResearch Sc   | 2              | Married     | 1             | No       |
| EM00136 | 26  | 841            | Research & Development | Other,Female          | 46      | 2       | 1        | RSResearch Sc   | 2              | Married     | 1             | No       |
| EM00137 | 58  | 147            | Research & Development | Medical,Female        | 94      | 3       | 3        | HCHHealthcare I | 4              | Married     | 1             | No       |
| EM00137 | 58  | 147            | Research & Development | Medical,Female        | 94      | 3       | 3        | HCHHealthcare I | 4              | Married     | 1             | No       |
| EM00138 | 19  | 528            | Sales                  | Marketing,Male        | 50      | 3       | 1        | SRSales Repres  | 3              | Single      | 1             | Yes      |
| EM00138 | 19  | 528            | Sales                  | Marketing,Male        | 50      | 3       | 1        | SRSales Repres  | 3              | Single      | 1             | Yes      |
| EM00139 | 22  | 594            | Research & Development | Technical Degree,Male | 100     | 3       | 1        | LTLaboratory Te | 4              | Married     | 0             | No       |
| EM00139 | 22  | 594            | Research & Development | Technical Degree,Male | 100     | 3       | 1        | LTLaboratory Te | 4              | Married     | 0             | No       |
| EM00140 | 49  | 470            | Research & Development | Medical,Female        | 96      | 3       | 2        | MDManufacturi   | 1              | Married     | 1             | No       |
| EM00140 | 49  | 470            | Research & Development | Medical,Female        | 96      | 3       | 2        | MDManufacturi   | 1              | Married     | 1             | No       |
| EM00141 | 43  | 957            | Research & Development | Medical,Female        | 72      | 4       | 1        | RSResearch Sc   | 3              | Single      | 4             | No       |
| EM00141 | 43  | 957            | Research & Development | Medical,Female        | 72      | 4       | 1        | RSResearch Sc   | 3              | Single      | 4             | No       |
| EM00142 | 50  | 809            | Sales                  | Marketing,Female      | 77      | 3       | 3        | SESales Execut  | 4              | Single      | 4             | No       |
| EM00142 | 50  | 809            | Sales                  | Marketing,Female      | 77      | 3       | 3        | SESales Execut  | 4              | Single      | 4             | No       |

## After Removing Duplicates:

| EMPID    | Age | DailyRate (\$) | Department             | Education, Gender     | HourlyR | JobInvc | JobLevel | Prefix Job N    | JobSatisfactio | MaritalStat | NumCompaniesW | OverTime | TotalW |
|----------|-----|----------------|------------------------|-----------------------|---------|---------|----------|-----------------|----------------|-------------|---------------|----------|--------|
| EM00131  | 30  | 1312           | Research & Development | Life Sciences,Male    | 96      | 1       | 1        | RSResearch Sc   | 3              | Divorced    | 1             | No       | 10     |
| EM00132  | 33  | 750            | Sales                  | Marketing,Male        | 95      | 3       | 2        | SESales Execut  | 2              | Married     | 0             | No       | 8      |
| EM00133  | 56  | 441            | Research & Development | Life Sciences,Female  | 72      | 3       | 1        | RSResearch Sc   | 2              | Married     | 9             | Yes      | 7      |
| EM00134  | 51  | 684            | Research & Development | Life Sciences,Male    | 51      | 3       | 5        | RDRResearch Di  | 3              | Single      | 7             | No       | 23     |
| EM00135  | 31  | 249            | Sales                  | Life Sciences,Male    | 76      | 1       | 2        | SESales Execut  | 3              | Married     | 4             | Yes      | 12     |
| EM00136  | 26  | 841            | Research & Development | Other,Female          | 46      | 2       | 1        | RSResearch Sc   | 2              | Married     | 1             | No       | 5      |
| EM00137  | 58  | 147            | Research & Development | Medical,Female        | 94      | 3       | 3        | HCHHealthcare I | 4              | Married     | 1             | No       | 40     |
| EM00138  | 19  | 528            | Sales                  | Marketing,Male        | 50      | 3       | 1        | SRSales Repres  | 3              | Single      | 1             | Yes      | 0      |
| EM00139  | 22  | 594            | Research & Development | Technical Degree,Male | 100     | 3       | 1        | LTLaboratory Te | 4              | Married     | 0             | No       | 3      |
| EM00140  | 49  | 470            | Research & Development | Medical,Female        | 96      | 3       | 2        | MDManufacturi   | 1              | Married     | 1             | No       | 16     |
| EM00141  | 43  | 957            | Research & Development | Medical,Female        | 72      | 4       | 1        | RSResearch Sc   | 3              | Single      | 4             | No       | 18     |
| EM00142  | 50  | 809            | Sales                  | Marketing,Female      | 77      | 3       | 3        | SESales Execut  | 4              | Single      | 4             | No       | 16     |
| EM00100  | 46  | 669            | Sales                  | Medical,Male          | 64      | 2       | 3        | SESales Execut  | 4              | Single      | 1             | No       | 9      |
| EM001000 | 37  | 1439           | Research & Development | Life Sciences,Male    | 54      | 3       | 1        | RSResearch Sc   | 3              | Married     | 7             | Yes      | 8      |
| EM001001 | 33  | 1111           | Sales                  | Life Sciences,Male    | 61      | 3       | 2        | SESales Execut  | 4              | Married     | 6             | No       | 8      |
| EM001002 | 32  | 499            | Sales                  | Marketing,Male        | 36      | 3       | 2        | SESales Execut  | 2              | Married     | 0             | Yes      | 4      |
| EM001003 | 39  | 1485           | Research & Development | Life Sciences,Male    | 71      | 3       | 3        | HCHHealthcare I | 3              | Married     | 3             | No       | 13     |

Microsoft Excel

12 duplicate values found and removed: 1453 unique values remain. Note that counts may include empty cells, spaces, etc.

OK

## ❖ Your client needs analysis by gender and education level.

- Which column can you use for analysis on this? Perform data cleaning for the column you have identified.

|    | DailyRate (\$) | Department                  | Education, Gender     | HourlyR | JobInvc | JobLevel | Pre |
|----|----------------|-----------------------------|-----------------------|---------|---------|----------|-----|
| 30 |                | 1312 Research & Development | Life Sciences,Male    | 96      | 1       | 1        | RSI |
| 33 |                | 750 Sales                   | Marketing,Male        | 95      | 3       | 2        | SE  |
| 56 |                | 441 Research & Development  | Life Sciences,Female  | 72      | 3       | 1        | RSI |
| 51 |                | 684 Research & Development  | Life Sciences,Male    | 51      | 3       | 5        | RD  |
| 31 |                | 249 Sales                   | Life Sciences,Male    | 76      | 1       | 2        | SE  |
| 26 |                | 841 Research & Development  | Other,Female          | 46      | 2       | 1        | RSI |
| 58 |                | 147 Research & Development  | Medical,Female        | 94      | 3       | 3        | HC  |
| 19 |                | 528 Sales                   | Marketing,Male        | 50      | 3       | 1        | SR  |
| 22 |                | 594 Research & Development  | Technical Degree,Male | 100     | 3       | 1        | LT  |

"The problem in this Column was there are two variables in one column, so we have to separate these two variables by using Text to Column tool with a comma delimiter."

## After Cleaning:

| Age | DailyRate (\$) | Department             | Education        | Gender | HourlyR | JobInvc | JobLevel | Prefix_Job R    | JobSatisfaction | MaritalStat | NumCompaniesW |
|-----|----------------|------------------------|------------------|--------|---------|---------|----------|-----------------|-----------------|-------------|---------------|
| 30  | 1312           | Research & Development | Life Sciences    | Male   | 96      | 1       | 1        | RSResearch Sci  | 3               | Divorced    |               |
| 33  | 750            | Sales                  | Marketing        | Male   | 95      | 3       | 2        | SESales Executi | 2               | Married     |               |
| 56  | 441            | Research & Development | Life Sciences    | Female | 72      | 3       | 1        | RSResearch Sci  | 2               | Married     |               |
| 51  | 684            | Research & Development | Life Sciences    | Male   | 51      | 3       | 5        | RDRResearch Dir | 3               | Single      |               |
| 31  | 249            | Sales                  | Life Sciences    | Male   | 76      | 1       | 2        | SESales Executi | 3               | Married     |               |
| 26  | 841            | Research & Development | Other            | Female | 46      | 2       | 1        | RSResearch Sci  | 2               | Married     |               |
| 58  | 147            | Research & Development | Medical          | Female | 94      | 3       | 3        | HCHHealthcare f | 4               | Married     |               |
| 19  | 528            | Sales                  | Marketing        | Male   | 50      | 3       | 1        | SRSales Repres  | 3               | Single      |               |
| 22  | 594            | Research & Development | Technical Degree | Male   | 100     | 3       | 1        | LTLaboratory Te | 4               | Married     |               |
| 49  | 470            | Research & Development | Medical          | Female | 96      | 3       | 2        | MDManufacturi   | 1               | Married     |               |
| 43  | 957            | Research & Development | Medical          | Female | 72      | 4       | 1        | RSResearch Sci  | 3               | Single      |               |
| 50  | 809            | Sales                  | Marketing        | Female | 77      | 3       | 3        | SESales Executi | 4               | Single      |               |
| 46  | 669            | Sales                  | Medical          | Male   | 64      | 2       | 3        | SESales Executi | 4               | Single      |               |
| 37  | 1439           | Research & Development | Life Sciences    | Male   | 54      | 3       | 1        | RSResearch Sci  | 3               | Married     |               |
| 33  | 1111           | Sales                  | Life Sciences    | Male   | 61      | 3       | 2        | SESales Executi | 4               | Married     |               |
| 32  | 499            | Sales                  | Marketing        | Male   | 36      | 3       | 2        | SESales Executi | 2               | Married     |               |
| 39  | 1485           | Research & Development | Life Sciences    | Male   | 71      | 3       | 3        | HCHHealthcare f | 3               | Married     |               |
| 25  | 1372           | Sales                  | Life Sciences    | Male   | 93      | 4       | 2        | SESales Executi | 3               | Married     |               |
| 52  | 322            | Research & Development | Medical          | Female | 59      | 4       | 4        | MDManufacturi   | 3               | Married     |               |
| 43  | 930            | Research & Development | Medical          | Female | 73      | 2       | 2        | RSResearch Sci  | 3               | Single      |               |
| 27  | 205            | Sales                  | Marketing        | Female | 98      | 2       | 2        | SESales Executi | 4               | Married     |               |

❖ Separate the prefix and job role in two separate columns.

| HourlyR | JobInvc | JobLevel | Prefix_Job Role              | JobSatisfaction | MaritalStat | NumCompanies |
|---------|---------|----------|------------------------------|-----------------|-------------|--------------|
| 96      | 1       | 1        | RSResearch Scientist         | 3               | Divorced    |              |
| 95      | 3       | 2        | SESales Executive            | 2               | Married     |              |
| 72      | 3       | 1        | RSResearch Scientist         | 2               | Married     |              |
| 51      | 3       | 5        | RDRResearch Director         | 3               | Single      |              |
| 76      | 1       | 2        | SESales Executive            | 3               | Married     |              |
| 46      | 2       | 1        | RSResearch Scientist         | 2               | Married     |              |
| 94      | 3       | 3        | HCHHealthcare Representative | 4               | Married     |              |
| 50      | 3       | 1        | SRSales Representative       | 3               | Single      |              |
| 100     | 3       | 1        | LTLaboratory Technician      | 4               | Married     |              |
| 96      | 3       | 2        | MDManufacturing Director     | 1               | Married     |              |
| 72      | 4       | 1        | RSResearch Scientist         | 3               | Single      |              |

## After Separate:

- =LEFT([@[Prefix\_Job Role]],2), Use For separation of Prefix
- =RIGHT([@[Prefix\_Job Role]],LEN([@[Prefix\_Job Role]])-2), use for separation of job role

AutoSave

Off

Employee Data Assignment Compl...

Search

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ClipboardFontAlignmentNumberStylesCellsEditingSensitivityAdd-ins

L3

=RIGHT([@Prefix\_Job Role],LEN([@Prefix\_Job Role])-2)

|    | F      | G          | H              | I        | J                           | K      | L                         | M               | N             | O                  |          |
|----|--------|------------|----------------|----------|-----------------------------|--------|---------------------------|-----------------|---------------|--------------------|----------|
|    | Gender | HourlyRate | JobInvolvement | JobLevel | Prefix_Job Role             | Prefix | Job Role                  | JobSatisfaction | MaritalStatus | NumCompaniesWorked | OverTime |
| 1  | Male   | 64         | 2              | 3        | SESales Executive           | SE     | Sales Executive           | 4               | Single        |                    | 1 No     |
| 2  | Male   | 54         | 3              | 1        | RSResearch Scientist        | RS     | Research Scientist        | 3               | Married       |                    | 7 Yes    |
| 3  | Male   | 61         | 3              | 2        | SESales Executive           | SE     | Sales Executive           | 4               | Married       |                    | 6 No     |
| 4  | Male   | 36         | 3              | 2        | SESales Executive           | SE     | Sales Executive           | 2               | Married       |                    | 0 Yes    |
| 5  | Male   | 71         | 3              | 3        | HCHealthcare Representative | HC     | Healthcare Representative | 3               | Married       |                    | 3 No     |
| 6  | Male   | 93         | 4              | 2        | SESales Executive           | SE     | Sales Executive           | 3               | Married       |                    | 2 No     |
| 7  | Female | 59         | 4              | 4        | MDManufacturing Director    | MD     | Manufacturing Director    | 3               | Married       |                    | 2 Yes    |
| 8  | Female | 73         | 2              | 2        | RSResearch Scientist        | RS     | Research Scientist        | 3               | Single        |                    | 1 Yes    |
| 9  | Female | 98         | 2              | 2        | SESales Executive           | SE     | Sales Executive           | 4               | Married       |                    | 1 Yes    |
| 10 | Female | 51         | 3              | 1        | RSResearch Scientist        | RS     | Research Scientist        | 3               | Single        |                    | 1 Yes    |
| 11 | Female | 36         | 2              | 1        | RSResearch Scientist        | RS     | Research Scientist        | 4               | Single        |                    | 0 No     |
| 12 | Male   | 78         | 2              | 4        | HCHealthcare Representative | HC     | Healthcare Representative | 2               | Married       |                    | 1 No     |
| 13 | Female | 31         | 3              | 4        | AMManager                   | AM     | Manager                   | 1               | Married       |                    | 0 No     |
| 14 | Female | 54         | 3              | 1        | LTLaboratory Technician     | LT     | Laboratory Technician     | 1               | Married       |                    | 9 No     |
| 15 | Female | 94         | 3              | 1        | LTLaboratory Technician     | LT     | Laboratory Technician     | 3               | Single        |                    | 4 No     |
| 16 | Male   | 60         | 3              | 3        | MDManufacturing Director    | MD     | Manufacturing Director    | 4               | Single        |                    | 2 No     |
| 17 | Male   | 81         | 3              | 1        | LTLaboratory Technician     | LT     | Laboratory Technician     | 4               | Married       |                    | 4 No     |
| 18 | Male   | 100        | 4              | 1        | LTLaboratory Technician     | LT     | Laboratory Technician     | 1               | Single        |                    | 0 No     |
| 19 | Male   | 54         | 2              | 2        | HRHuman Resources           | HR     | Human Resources           | 1               | Single        |                    | 1 No     |

## 2. Quick Analysis:

- ❖ Find total employees in sales department
- ❖ Find total employees in Research & Development
- ❖ Find total female employees

Formula for finding these:

- =COUNTIF(D:D,"Sales")
- =COUNTIF(D:D,"Research & Development ")
- =COUNTIF(F:F,"Female")

AutoSave Off Employee Data Assignment Compl... Search

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard Font Alignment Number

C10

|    | A | B                                          | C | D   | E | F | G |
|----|---|--------------------------------------------|---|-----|---|---|---|
| 1  |   |                                            |   |     |   |   |   |
| 2  |   |                                            |   |     |   |   |   |
| 3  |   |                                            |   |     |   |   |   |
| 4  |   | Total employees in Sales:                  |   | 439 |   |   |   |
| 5  |   | Total employees in Research & Development: |   | 951 |   |   |   |
| 6  |   | Total Employees in Human Resources:        |   | 63  |   |   |   |
| 7  |   | Total Female Employees:                    |   | 580 |   |   |   |
| 8  |   |                                            |   |     |   |   |   |
| 9  |   |                                            |   |     |   |   |   |
| 10 |   |                                            |   |     |   |   |   |

### 3. Data Transformation:

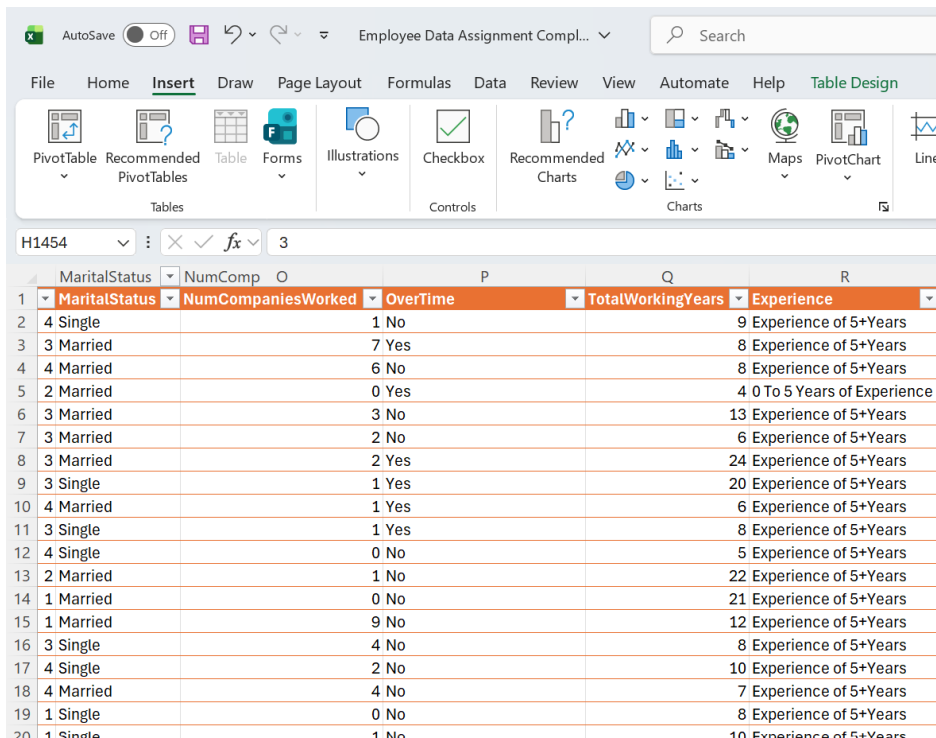
❖ Create a column to show employees who have worked for more than 5 years:

○ Label the values such that:

- If employee has worked more than 5 years, say “experience of 5+ years”, else say “0 to 5 years experience”

#### After Transforming the Data:

- =IF([@TotalWorkingYears]<5,"0 To 5 Years of Experience","Experience of 5+Years")  
Formula for this Transformation.



|    | MaritalStatus | NumComp            | O        | P                 | Q          | R                          |
|----|---------------|--------------------|----------|-------------------|------------|----------------------------|
|    | MaritalStatus | NumCompaniesWorked | OverTime | TotalWorkingYears | Experience |                            |
| 1  | 4 Single      |                    | 1 No     |                   | 9          | Experience of 5+Years      |
| 2  | 3 Married     |                    | 7 Yes    |                   | 8          | Experience of 5+Years      |
| 3  | 4 Married     |                    | 6 No     |                   | 8          | Experience of 5+Years      |
| 4  | 2 Married     |                    | 0 Yes    |                   | 4          | 0 To 5 Years of Experience |
| 5  | 3 Married     |                    | 3 No     |                   | 13         | Experience of 5+Years      |
| 6  | 3 Married     |                    | 2 No     |                   | 6          | Experience of 5+Years      |
| 7  | 3 Married     |                    | 2 Yes    |                   | 24         | Experience of 5+Years      |
| 8  | 3 Single      |                    | 1 Yes    |                   | 20         | Experience of 5+Years      |
| 9  | 4 Married     |                    | 1 Yes    |                   | 6          | Experience of 5+Years      |
| 10 | 3 Single      |                    | 1 Yes    |                   | 8          | Experience of 5+Years      |
| 11 | 4 Single      |                    | 0 No     |                   | 5          | Experience of 5+Years      |
| 12 | 2 Married     |                    | 1 No     |                   | 22         | Experience of 5+Years      |
| 13 | 1 Married     |                    | 0 No     |                   | 21         | Experience of 5+Years      |
| 14 | 1 Married     |                    | 9 No     |                   | 12         | Experience of 5+Years      |
| 15 | 3 Single      |                    | 4 No     |                   | 8          | Experience of 5+Years      |
| 16 | 4 Single      |                    | 2 No     |                   | 10         | Experience of 5+Years      |
| 17 | 4 Married     |                    | 4 No     |                   | 7          | Experience of 5+Years      |
| 18 | 1 Single      |                    | 0 No     |                   | 8          | Experience of 5+Years      |
| 19 | 1 Single      |                    | 1 No     |                   | 10         | Experience of 5+Years      |

❖ If employee has total working years as more than 15 and has worked in only one company, then say “Low risk employee” else says “high risk employee”

- =IF([@NumCompaniesWorked] = 0,"Data Error",  
IF(AND([@TotalWorkingYears]>15,[@NumCompaniesWorked]=1),"Low Risk Employee",  
"High Risk Employee")

Employee Data Assignment Compl...

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design

PivotTable Recommended PivotTables Tables

Formulas

Illustrations

Controls

Recommended Charts

Charts

Maps

PivotChart

Line Column Sparklines

S2 =IF([@NumCompaniesWorked] = 0,"Data Error", IF(AND([@TotalWorkingYears]>15,[@N

|    | N        | O                  | P        | Q                 | R                            | S                  |
|----|----------|--------------------|----------|-------------------|------------------------------|--------------------|
|    | alStatus | NumCompaniesWorked | OverTime | TotalWorkingYears | Experience                   | Level Of Risk      |
| 2  |          |                    | 1 No     |                   | 9 Experience of 5+Years      | High Risk Employee |
| 3  | ed       |                    | 7 Yes    |                   | 8 Experience of 5+Years      | High Risk Employee |
| 4  | ed       |                    | 6 No     |                   | 8 Experience of 5+Years      | High Risk Employee |
| 5  | ed       |                    | 0 Yes    |                   | 4 0 To 5 Years of Experience | Data Error         |
| 6  | ed       |                    | 3 No     |                   | 13 Experience of 5+Years     | High Risk Employee |
| 7  | ed       |                    | 2 No     |                   | 6 Experience of 5+Years      | High Risk Employee |
| 8  | ed       |                    | 2 Yes    |                   | 24 Experience of 5+Years     | High Risk Employee |
| 9  |          |                    | 1 Yes    |                   | 20 Experience of 5+Years     | Low Risk Employee  |
| 10 | ed       |                    | 1 Yes    |                   | 6 Experience of 5+Years      | High Risk Employee |
| 11 |          |                    | 1 Yes    |                   | 8 Experience of 5+Years      | High Risk Employee |
| 12 |          |                    | 0 No     |                   | 5 Experience of 5+Years      | Data Error         |
| 13 | ed       |                    | 1 No     |                   | 22 Experience of 5+Years     | Low Risk Employee  |
| 14 | ed       |                    | 0 No     |                   | 21 Experience of 5+Years     | Data Error         |
| 15 | ed       |                    | 9 No     |                   | 12 Experience of 5+Years     | High Risk Employee |
| 16 |          |                    | 4 No     |                   | 8 Experience of 5+Years      | High Risk Employee |
| 17 |          |                    | 2 No     |                   | 10 Experience of 5+Years     | High Risk Employee |
| 18 | ed       |                    | 4 No     |                   | 7 Experience of 5+Years      | High Risk Employee |
| 19 |          |                    | 0 No     |                   | 8 Experience of 5+Years      | Data Error         |
| 20 |          |                    | 1 No     |                   | 10 Experience of 5+Years     | High Risk Employee |

In this Transformation, I encountered records with **15+ years of experience**, but **0 companies worked**, which is logically impossible. While I could have **imputed** these values by changing the 0 to 1, I chose the **safer analytical approach** by using a **Nested IF** formula to label them as a **'Data Error'**. This ensures that invalid data does not pollute the 'High Risk' or 'Low Risk' categories, providing a more accurate final report.

#### 4. Reflection:

- ❖ Do you think any column is missing in this data which would have been helpful in analysing trends? If yes, which column could that be and how would you use it?
- ✓ A significant improvement to this data would be adding **'Date of Joining'** and **'Date of Leaving'**. I would use these to calculate the actual Attrition Rate for each department. This would allow the client to see if 'High Risk' employees are leaving due to recent changes in the Sales or R&D departments or if it is a long-term trend."